DEPI GRADUATION PROJECT

Task 4 - System Analysis & Design

Automated Teacher Workflow

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This document reflects the implemented solution: Google Forms for teacher input, Google Sheets for response capture/logging, n8n Cloud for orchestration, Google Gemini for AI generation, Google Docs for the editable lesson plan, and Gmail for clarification/delivery.

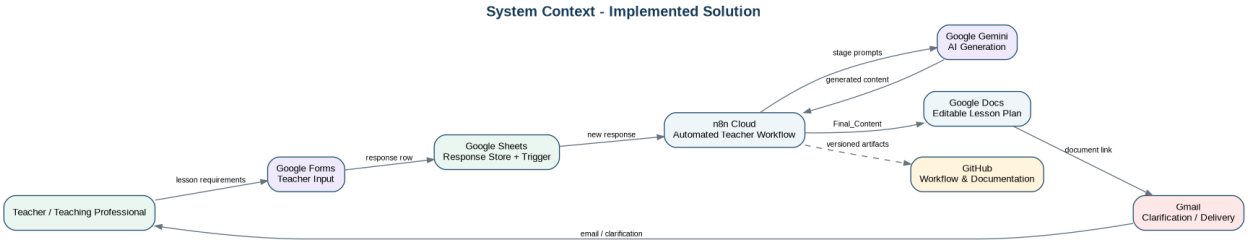


Figure 1. System Context Diagram - implemented Google Workspace, n8n Cloud, Gemini, Docs, Gmail, and repository context.

1. Problem Statement & Objectives

1.1 Problem Statement

The Automated Teacher Workflow streamlines lesson preparation by collecting teacher requirements through Google Forms, storing the response in Google Sheets, and using an n8n Cloud workflow to prepare and validate the input. Google Gemini generates the lesson outline, student activities, assessment, answer key, and differentiated material when requested. The workflow then merges the outputs into Final_Content, creates an editable Google Docs lesson plan, and supports teacher communication through Gmail. This reduces repetitive planning work while keeping the teacher responsible for reviewing the final material.

1.2 Project Objectives

- Collect structured lesson requirements through Google Forms and capture them in Google Sheets.
- Automatically trigger the n8n Cloud workflow when a new form response is available.
- Validate and prepare the submitted lesson data before AI generation begins.
- Use Google Gemini to generate the lesson outline, student activities, assessment, answer key, and differentiated material when requested.
- Merge the generated sections into one Final_Content field and create an editable Google Docs lesson plan.
- Use Gmail to request clarification for missing required inputs and to support delivery of the generated document link.

1.3 System Boundary

In Scope	Out of Scope
Google Forms teacher input; Google Sheets response storage/trigger; n8n Cloud orchestration; Google Gemini content generation; IF validation and differentiation branching; Merge/Edit Fields; Google Docs document creation; Gmail clarification/delivery; Google Sheets logging.	Full LMS; direct student-facing application; automatic grading of student submissions; training or fine-tuning a custom AI model.
Editable lesson-plan output; teacher review before use; workflow documentation, diagrams, prompts, and testing evidence.	Automatic classroom publishing or use of generated content without teacher review.

2. Use Case Diagram & Descriptions

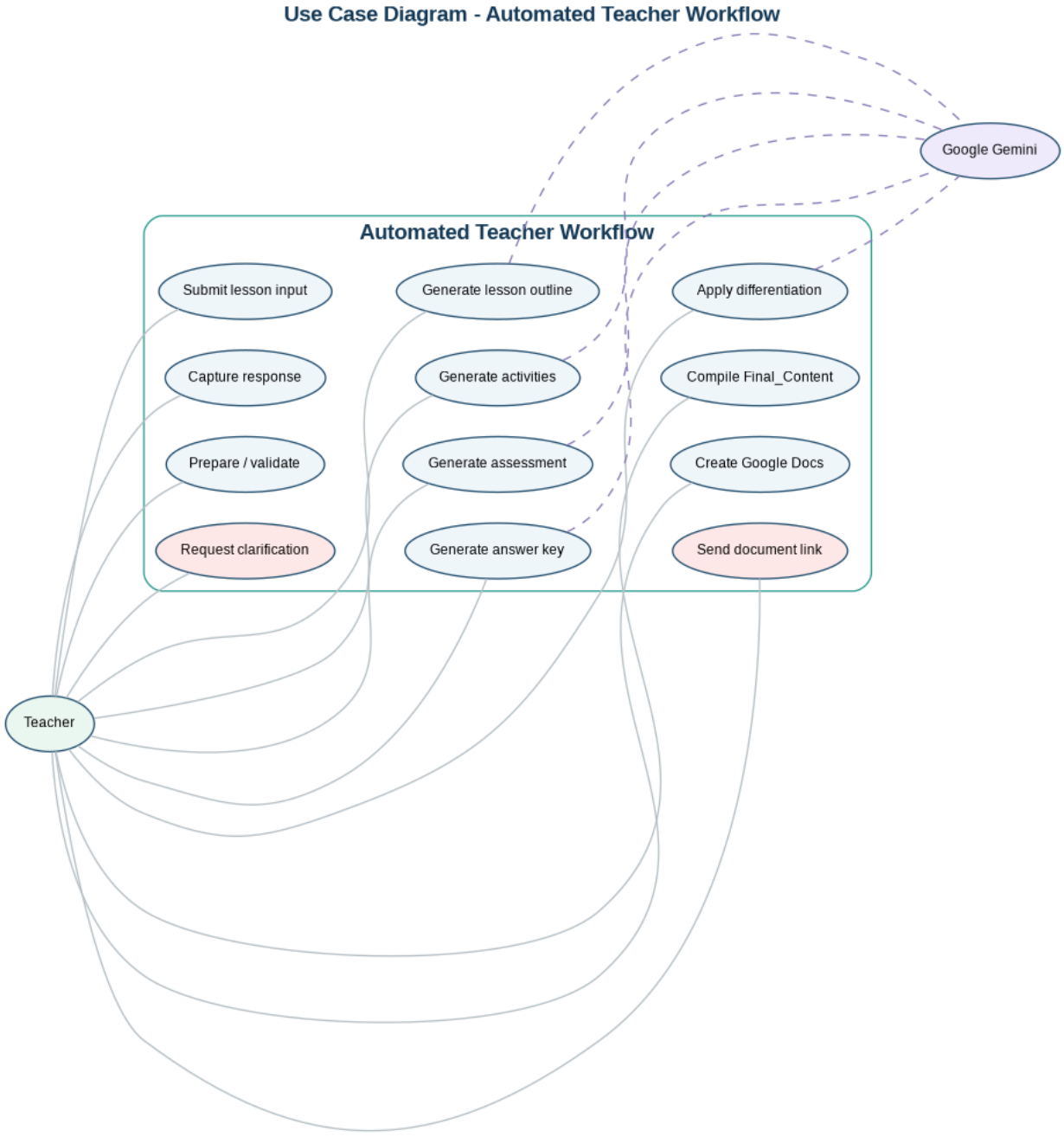


Figure 2. Use Case Diagram - teacher interactions and implemented workflow services.

ID	Use Case	Actor	Description	Result
UC-01	Submit Instructional Input	Teacher	Submit subject, grade level, topic, learning objective, lesson duration, and differentiation preference through Google Forms.	Form response is stored in Google Sheets.
UC-02	Capture Response & Trigger	System	Google Sheets receives the form response and the Google Sheets Trigger starts the n8n workflow.	New response enters n8n.
UC-03	Prepare & Validate Input	System	Edit Fields standardizes the values and an IF node checks required fields.	Valid input continues; invalid input is routed for clarification.
UC-04	Request Clarification	System / Teacher	Gmail sends a clarification request when required fields are missing.	Teacher receives a correction request.
UC-05	Generate Lesson Outline	AI Workflow	Google Gemini generates a structured lesson outline from the prepared context.	Lesson outline produced.
UC-06	Generate Student Activities	AI Workflow	Google Gemini generates activities aligned with the lesson requirements.	Student activities produced.
UC-07	Generate Assessment	AI Workflow	Google Gemini generates assessment questions aligned with the lesson context.	Assessment produced.
UC-08	Generate Answer Key	AI Workflow	Google Gemini generates answers corresponding to the assessment.	Answer key produced.
UC-09	Apply Differentiation	Workflow Branch / AI	An IF node checks the teacher preference; Google Gemini generates differentiated material when requested.	Differentiated material included when applicable.
UC-10	Compile Final Content	System	Merge combines the relevant outputs and Edit Fields builds the Final_Content field.	One combined lesson package is prepared.
UC-11	Create Google Docs Lesson Plan	System	Google Docs creates an editable lesson-planning document containing Final_Content.	Editable lesson document created.
UC-12	Deliver Document Link	System / Teacher	Gmail sends or is configured to send the generated document link and summary to the teacher.	Teacher can access the generated document.

3. Functional & Non-Functional Requirements

3.1 Functional Requirements

ID	Functional Requirement
FR-01	Collect teacher lesson information through Google Forms.
FR-02	Store the submitted form response in Google Sheets and use the response to trigger the n8n workflow.
FR-03	Use Edit Fields to standardize and prepare the submitted values for downstream nodes.
FR-04	Validate required input fields with an IF node before AI generation begins.
FR-05	Send a Gmail clarification request when mandatory input is missing.
FR-06	Generate a lesson outline using Google Gemini.
FR-07	Generate student activities using Google Gemini.
FR-08	Generate assessment questions using Google Gemini.
FR-09	Generate a corresponding answer key using Google Gemini.
FR-10	Use an IF branch to determine whether differentiated material is required.
FR-11	Generate a differentiated version with Google Gemini when the teacher requests it.
FR-12	Merge the relevant generated outputs into a single workflow path.

ID	Functional Requirement
FR-13	Assemble the combined sections into one Final_Content field using Edit Fields.
FR-14	Create an editable Google Docs lesson-plan document from Final_Content.
FR-15	Append/store teacher input data in Google Sheets for record/logging purposes.
FR-16	Support Gmail delivery of the generated Google Docs link to the teacher.
FR-17	Maintain the workflow and supporting project documentation in the project repository.

3.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	The teacher input and final lesson plan must be clearly structured and understandable without programming knowledge.
NFR-02	Reliability	The workflow should complete successfully for valid form submissions and handle invalid input through the clarification path.
NFR-03	Completeness	A successful run should generate the required lesson sections and combine them into Final_Content.
NFR-04	Branch Accuracy	The differentiation IF node should select the correct path based on the teacher preference.
NFR-05	Error Handling	Missing required input should not continue to AI generation; it should produce a clear clarification message.
NFR-06	Content Relevance	Generated content should remain aligned with subject, grade level, topic, learning objective, duration, and differentiation preference.
NFR-07	Editability	The generated lesson plan should remain editable in Google Docs so the teacher can review and adapt it.
NFR-08	Security	Google and AI credentials must be stored in n8n credentials/configuration and not embedded in shared workflow files or documentation.
NFR-09	Privacy	The workflow should process only the lesson information required for generation and avoid unnecessary personal/student data.
NFR-10	Maintainability	Workflow nodes, prompts, field names, and documentation should be organized and consistently named for future updates.
NFR-11	Integration	The n8n workflow should use supported Google Workspace and Google Gemini connectors available in the project environment.

4. Software Architecture

The implemented solution uses a cloud-based workflow architecture. Google Forms collects teacher lesson requirements and Google Sheets stores the response. A Google Sheets Trigger starts the n8n Cloud workflow, where Edit Fields prepares the values and IF nodes control validation and differentiation. Google Gemini Chat Model nodes generate the educational content. Merge and Edit Fields assemble Final_Content, Google Docs creates the editable lesson plan, and Gmail supports clarification and document-link delivery.

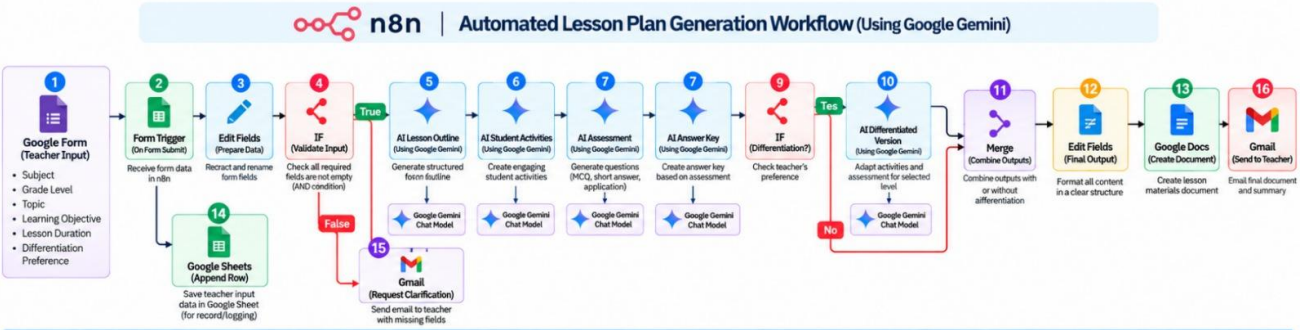


Figure 3. Implemented n8n workflow overview supplied by the project team.

Layer	Component	Responsibility
Presentation / Input	Google Forms	Collect teacher lesson requirements.

Layer	Component	Responsibility
Data Capture / Trigger	Google Sheets + Google Sheets Trigger	Store form responses and start the n8n workflow when a new response is available.
Workflow Orchestration	n8n Cloud	Prepare data, validate inputs, route conditions, call AI nodes, merge outputs, and coordinate integrations.
AI Generation	Google Gemini Chat Model	Generate lesson outline, student activities, assessment, answer key, and differentiated material.
Output / Document	Google Docs	Create and store the combined editable lesson-plan document.
Notification	Gmail	Request clarification for missing fields and support delivery of the generated document link.
Version Control	GitHub / project repository	Store workflow exports, prompts, diagrams, documentation, and revisions.

5. Database Design & Data Modeling

The implemented workflow does not use a separate relational database. Google Sheets acts as the lightweight response and logging store, while n8n carries prepared fields and AI outputs as runtime JSON/item data. The final combined content is stored as an editable Google Docs document. The conceptual data model below documents the relationships among these implemented data objects without claiming a relational database that is not part of the current solution.

5.1 ER Diagram

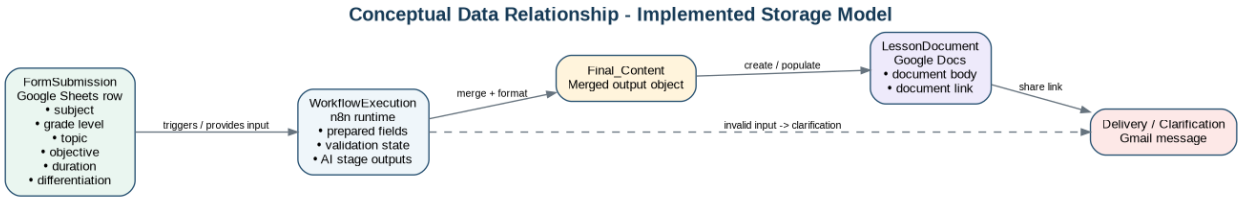


Figure 4. Conceptual Data Relationship Diagram - Google Sheets response, n8n runtime, Final_Content, Google Docs, and Gmail.

5.2 Logical Schema

Data Object	Key Fields / Content	Storage / Purpose
FormSubmission	subject; grade_level; topic; learning_objective; lesson_duration; differentiation_preference; response timestamp	Google Sheets response row; source data for the workflow.
PreparedInput	standardized field names and cleaned values	n8n runtime item produced by Edit Fields.
AIStageOutput	lesson_outline; student_activities; assessment; answer_key; differentiated_version (when requested)	n8n runtime outputs returned by Google Gemini nodes.
FinalContent	Final_Content combined text	n8n merged/formatted payload used to populate Google Docs.
LessonDocument	document body; Google Docs document reference/link	Google Docs; editable teacher-facing lesson plan.
EmailMessage	clarification message or document-link delivery payload	Gmail; teacher communication.

5.3 Physical Schema

Physical Store / Structure	Implemented Fields / Representation	Keys & Constraints
Google Sheets response row	Subject; Grade Level; Topic; Learning Objective; Lesson Duration; Differentiation Preference; response timestamp	No relational PK/FK; one Google Forms response is stored as one sheet row.
n8n prepared item	Normalized lesson fields after Edit Fields	Field names are kept consistent across downstream nodes.
n8n AI outputs	Lesson outline; Student activities; Assessment; Answer key; Differentiated version when applicable	Outputs are generated as node data and passed to Merge.
Final_Content	Combined and formatted lesson sections created by Edit Fields1	Single final field used as the Google Docs document body.

Physical Store / Structure	Implemented Fields / Representation	Keys & Constraints
Google Docs document	Editable lesson-plan content + generated document link/reference	Created from Final_Content for teacher review.
Gmail payload	Clarification or delivery subject/body + teacher recipient + document link where applicable	Uses Gmail credential in n8n; exact recipient source depends on the form/workflow configuration.

5.4 Normalization Considerations

- One Google Forms response is represented by one Google Sheets row to keep the submitted lesson context together.
- Edit Fields standardizes names before AI processing so the same context can be reused across all Google Gemini stages.
- AI outputs remain separate until the Merge node, preventing one stage from overwriting another stage output.
- Final_Content is created only after the relevant branches are complete, then stored in Google Docs as the editable final artifact.

6. Data Flow & System Behavior

6.1 DFD - Context Level

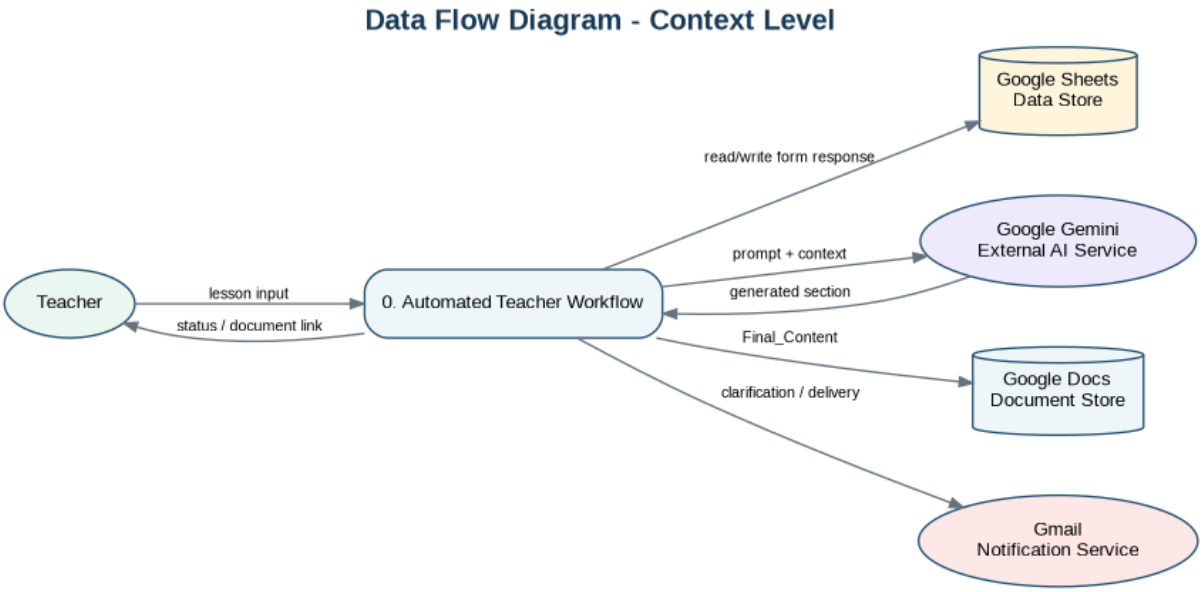


Figure 5. DFD Context Level - external services and teacher data flows.

6.2 DFD - Level 1

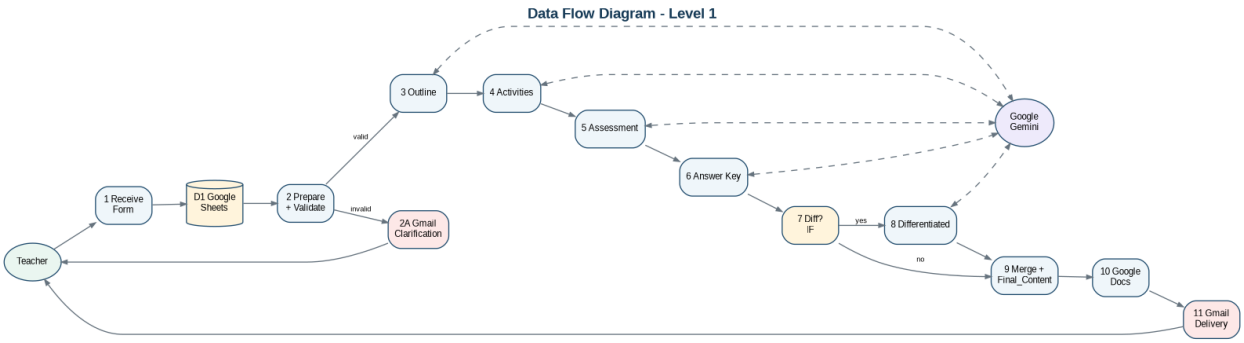


Figure 6. DFD Level 1 - implemented generation, validation, differentiation, merge, document, and delivery flow.

6.3 Sequence Diagram

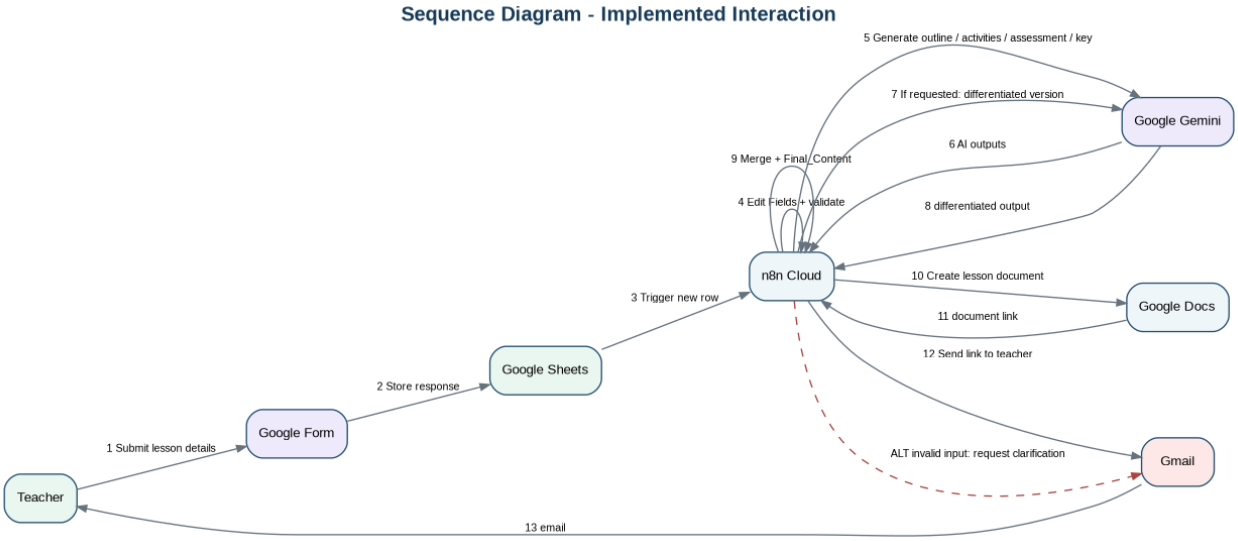


Figure 7. Sequence Diagram - Google Form submission through Google Docs and Gmail delivery.

6.4 Activity Diagram

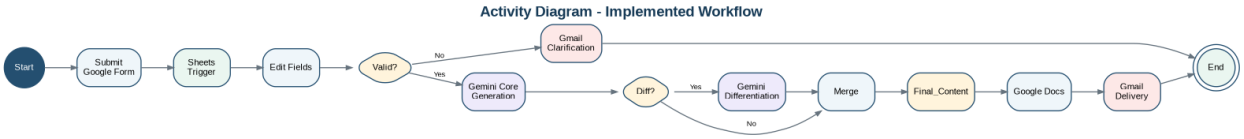


Figure 8. Activity Diagram - implemented validation, generation, branching, merge, and output flow.

6.5 State Diagram

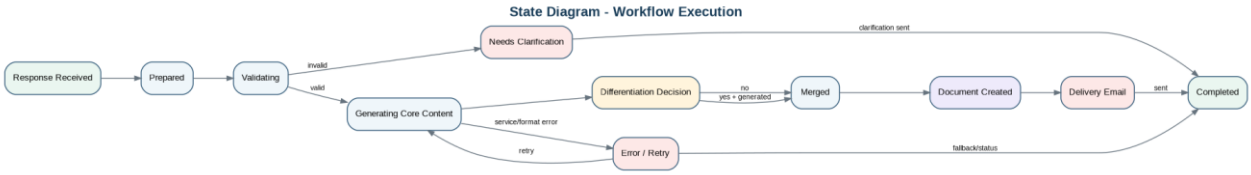


Figure 9. State Diagram - workflow execution lifecycle.

6.6 Class Diagram

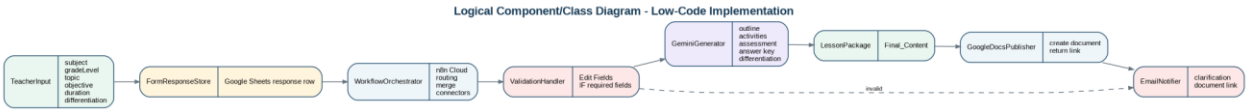


Figure 10. Logical Component/Class Diagram - low-code implementation responsibilities.

Logical Component	Responsibility
TeacherInput	Represents the lesson requirements submitted through Google Forms.
FormResponseStore	Stores the Google Forms response in Google Sheets and exposes it to the trigger.
WorkflowOrchestrator	Coordinates n8n nodes, routing, validation, AI calls, merge, and external services.
ValidationHandler	Uses Edit Fields and IF logic to prepare data and stop incomplete requests.
GeminiGenerator	Produces stage-specific lesson content using Google Gemini.
LessonPackage	Represents the merged Final_Content output.
GoogleDocsPublisher	Creates the editable lesson-plan document from Final_Content.
EmailNotifier	Sends clarification or document-link messages through Gmail.

7. UI/UX Design & Prototyping

7.1 Wireframes & Mockups

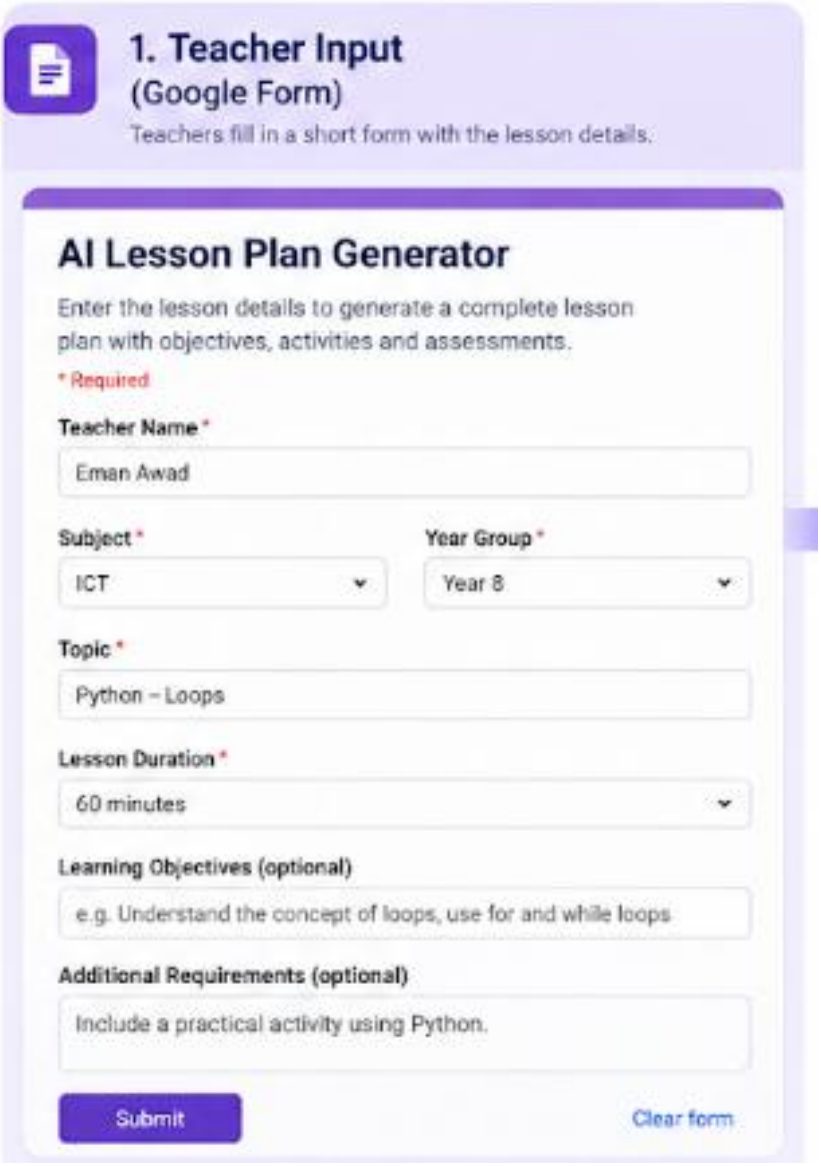


Figure 11. Teacher Input Form - Google Forms implementation visual supplied by the project team.



Figure 12. Teacher Review Output - generated Google Docs lesson-plan visual supplied by the project team.

7.2 UI/UX Guidelines

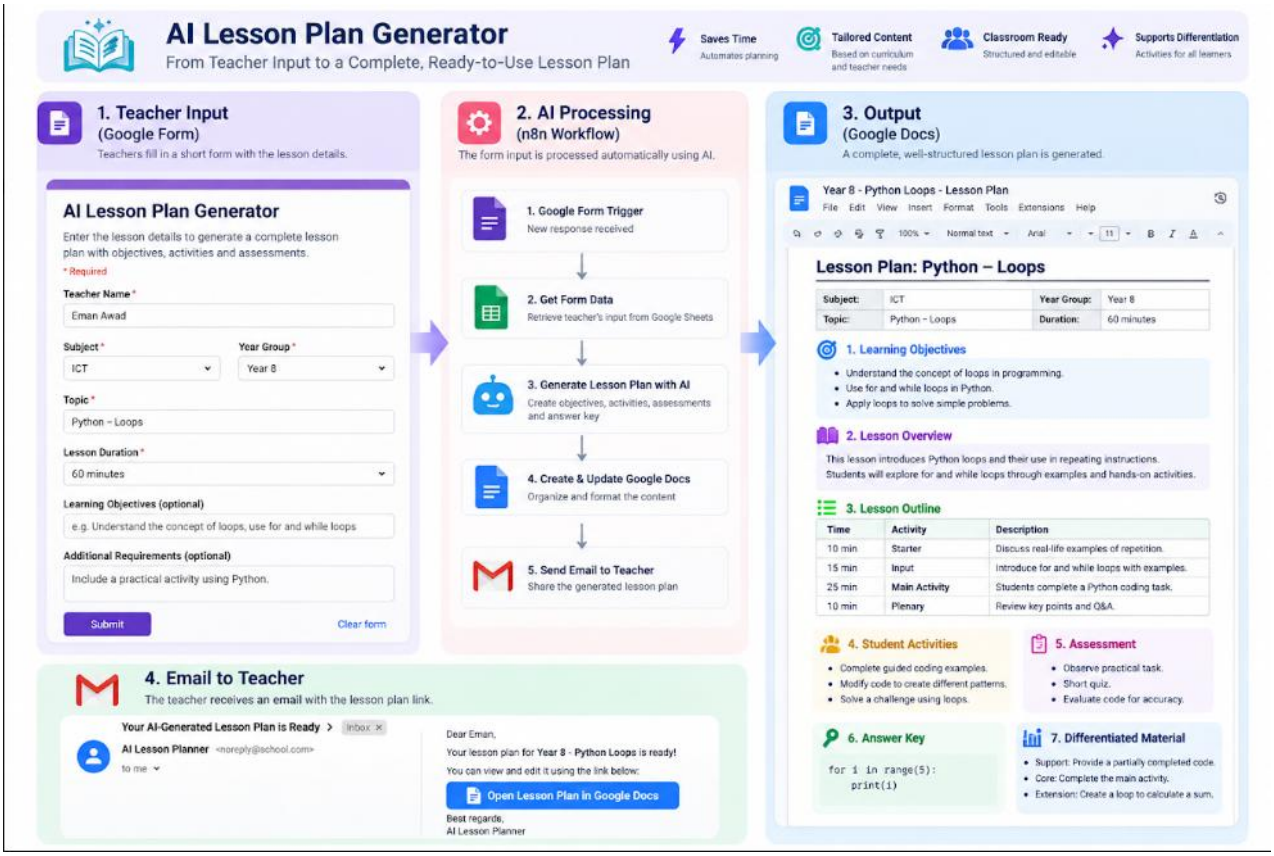


Figure 13. End-to-end teacher experience - Google Form, n8n/AI processing, Google Docs output, and email delivery.

Principle	Guideline
Clarity	Keep Google Form labels short and explicit for subject, grade level, topic, objective, duration, and differentiation preference.
Consistency	Use the same field names from Google Forms/Sheets through Edit Fields and AI prompts.
Validation	Route incomplete submissions to a specific clarification email instead of continuing generation.
Review-first	Present the output as an editable Google Docs draft that the teacher reviews before use.
Structure	Use a consistent lesson-plan hierarchy for outline, activities, assessment, answer key, and differentiated material.
Accessibility	Use readable text, clear headings, sufficient contrast, and avoid status communication through color alone.
Editability	Keep the final output in Google Docs so teachers can revise and adapt the generated content.
Feedback	Communicate missing-input or delivery status through clear Gmail messages where applicable.

8. System Deployment & Integration

8.1 Technology Stack

Layer	Technology	Purpose
Teacher Input	Google Forms	Collect subject, grade level, topic, learning objective, lesson duration, and differentiation preference.
Response Store / Trigger	Google Sheets + Google Sheets Trigger	Store form responses and trigger n8n on a new submission.
Backend / Orchestration	n8n Cloud	Run validation, AI stages, branching, merging, formatting, and service integrations.
AI Layer	Google Gemini Chat Model	Generate the lesson outline, activities, assessment, answer key, and differentiated content.

Layer	Technology	Purpose
Data Exchange	n8n item / JSON-style fields	Carry prepared values and generated outputs between nodes.
Output Layer	Google Docs	Create the combined editable lesson-plan document.
Notification	Gmail	Request missing information and support delivery of the document link.
Version Control	GitHub / project repository	Store workflow export, prompts, diagrams, documentation, and revisions.

8.2 Component Diagram

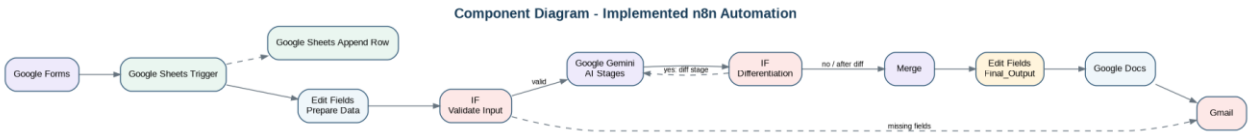


Figure 14. Component Diagram - implemented n8n nodes and connected Google services.

8.3 Deployment Diagram

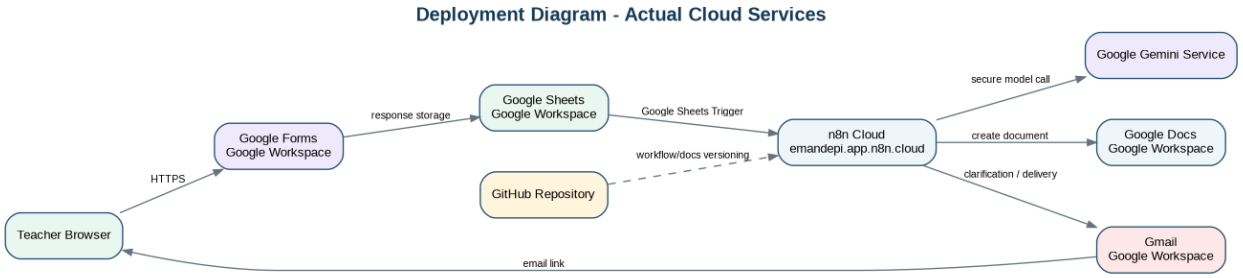


Figure 15. Deployment Diagram - n8n Cloud, Google Workspace services, Google Gemini, and repository.

8.4 Integration Security

- Store Google Workspace and Google Gemini credentials in n8n credentials or secure workspace configuration.
- Do not place API keys, OAuth tokens, or credentials in GitHub, screenshots, documentation, or exported examples.
- Send only the lesson-level information required by the relevant Google Gemini generation stage.
- Validate required input before AI calls and keep clarification/error routes separate from successful output delivery.
- Use the minimum required Google Sheets, Google Docs, and Gmail permissions for the connected project accounts.

9. Additional Deliverables

9.1 API Documentation

The implemented system uses n8n connectors for Google Sheets, Google Gemini, Google Docs, and Gmail. The connector-level design below reflects the supplied workflow documentation. The exact Google Gemini model variant is not specified in the provided implementation materials.

Connector / Integration	Implementation Design
Google Sheets Trigger	Reads a newly stored Google Forms response and starts the n8n workflow with the submitted lesson fields.
Google Sheets Append Row	Stores/records the teacher input data in Google Sheets for logging/record purposes.
Google Gemini Chat Model	Receives stage-specific prompts plus the prepared lesson context and returns generated text for outline, activities, assessment, answer key, and differentiation.
Google Docs	Creates the lesson-plan document and inserts Final_Content into the document body, returning a document reference/link.
Gmail	Sends a clarification request for invalid/missing input and supports sending the generated Google Docs link to the teacher.
Validation / Data Contract	Edit Fields normalizes values; IF nodes validate inputs and differentiation; downstream steps expect non-empty generated output before merge/document creation.

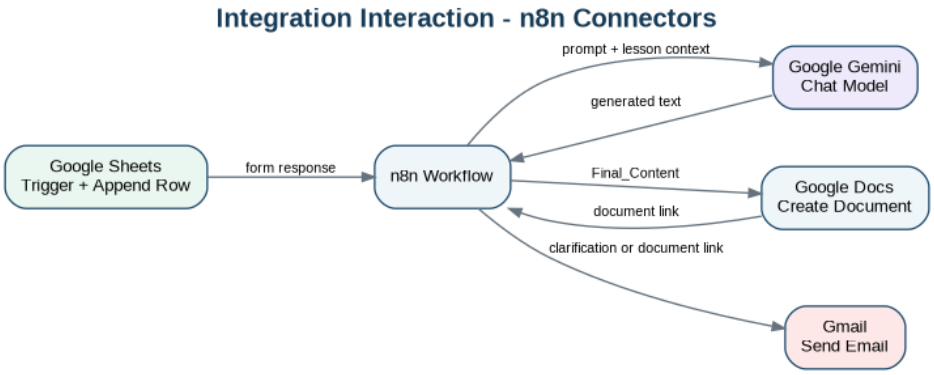


Figure 16. n8n Connector Interaction - Google Sheets, Google Gemini, Google Docs, and Gmail.

9.2 Testing & Validation

Testing should verify the implemented path from Google Forms and Google Sheets through n8n, Google Gemini, Google Docs, and Gmail. The project documentation defines the following success criteria: a new form response starts the workflow, required inputs reach the AI steps, expected lesson sections are combined into Final_Content, a Google Docs document is created and populated, and teacher delivery is complete after the Gmail node is configured and tested.

Test Level	Coverage	Pass Evidence
Node / Unit Testing	Edit Fields preparation; IF validation; differentiation IF; each Google Gemini stage; Merge; Final_Content formatting; Google Docs; Gmail.	Representative input produces the expected node result and required fields.
Integration Testing	Google Forms -> Google Sheets -> n8n -> Gemini -> Merge/Edit Fields -> Google Docs -> Gmail.	Data moves correctly through the complete workflow and the Google Docs document is populated.
User Acceptance Testing	Teacher reviews the Google Docs lesson plan for clarity, relevance, completeness, editability, and differentiation.	Teacher can open, review, and edit the generated document with acceptable corrections.
Failure / Recovery Testing	Missing required fields; invalid differentiation value; AI/service failure; Docs/Gmail connector failure.	Workflow stops or routes safely and provides a clear clarification/error status instead of a misleading final output.
Regression Testing	Re-run previously successful form submissions after prompt or workflow changes.	Existing core behavior remains functional after updates.

9.3 Deployment Strategy

Step	Deployment Activity
1	Connect the Google Form to the Google Sheets response spreadsheet used by the project.
2	Configure the n8n Cloud workflow in the team workspace and connect the Google Sheets Trigger to the response sheet.
3	Configure Google Gemini credentials/model nodes and the stage-specific prompts.
4	Configure Google Docs and Gmail credentials with the minimum required permissions.
5	Run validation, differentiation, end-to-end document-creation, and clarification/delivery test cases.
6	Activate/freeze the validated workflow version for the final demo and preserve the workflow export and documentation in the project repository.

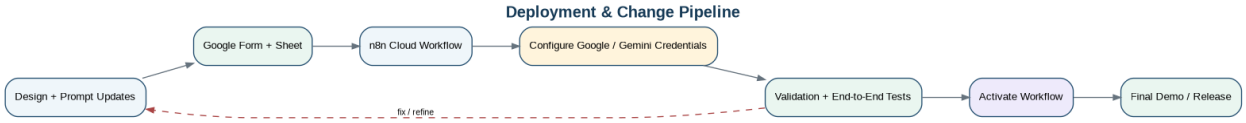


Figure 17. Deployment and change pipeline for the implemented cloud workflow.

10. Design Summary

The final system design now reflects the implemented Automated Teacher Workflow. Google Forms captures teacher input, Google Sheets stores the response and triggers n8n Cloud, Google Gemini generates the lesson sections, IF nodes manage validation and differentiation, Merge/Edit Fields create Final_Content, Google Docs produces the editable lesson plan, and Gmail supports clarification and document-link delivery. The design artifacts in this document are aligned with these implemented services rather than a separate custom application or relational database.

Design Area	Final Design Decision
Primary Actor	Teacher / teaching professional
Project Name	Automated Teacher Workflow
Input & Trigger	Google Forms -> Google Sheets -> Google Sheets Trigger in n8n
Core Platform	n8n Cloud
AI Generation	Google Gemini Chat Model
Core Outputs	Lesson outline; student activities; assessment; answer key; differentiated material when requested; combined Final_Content
Output & Delivery	Google Docs editable lesson plan; Gmail clarification/delivery support
Persistence / Logging	Google Sheets response/log rows; Google Docs final document; n8n execution data
Human Control	Teacher review and editing required before classroom use
Implementation Assets	Shared Google Drive folder containing the exported n8n workflow JSON and supporting project files. Open Shared Google Drive Folder